



Marketing Strategy

Objective: To analyse how paywall implementation (Freemium vs. Subscription Model) affects Engagement and Word of Mouth

ABSTRACT

This study examines the impact of paywalls on consumer behavior and business models within digital streaming, focusing on Spotify's freemium strategy. Through a quantitative survey, it explores user engagement, satisfaction, motivations for upgrading or downgrading, and the role of advertisements. Findings highlight how demographic factors, emotional attachment, perceived value, and technological infrastructure influence subscription decisions and user retention. Comparative analysis of content-only, ad-only, and combination models reveals key trade-offs in monetization strategies. The study underscores the importance of balancing ad intensity, pricing, and feature differentiation while leveraging AI and user data to optimize freemium models. Insights contribute to refining digital monetization frameworks and enhancing long-term user engagement in the streaming industry.

INTRODUCTION

The emergence of freemium and subscription-based models has radically altered the digital music streaming environment, with Spotify emerging as a leader in exploiting these techniques to increase user acquisition and engagement (Wagner, Benlian, & Hess, 2016). Spotify's dual-tier strategy, which includes a free, ad-supported tier and a premium, subscription-based service, has spurred debate on the psychological, behavioural, and technological factors that influence user pleasure and monetisation (Mäntymäki, Islam, and Benbasat, 2019). While boosting income possibilities, implementing a paywall offers a delicate balancing act: platforms must keep users engaged on the free tier while encouraging them towards premium upgrades (Guo et al., 2019). According to Oh, Animesh, and Pinsonneault (2016), personalisation and perceived control are important factors in moulding user experience and loyalty in freemium services. This is especially true for Spotify, where daily usage habits and suggestion satisfaction are critical to customer retention (Datta, Foubert, & Van Heerde, 2015).

Demographic issues hamper freemium conversion. Punj (2015) observes that desire to pay for digital entertainment changes with age, education, and income, implying that personalisation methods and pricing models should be customised accordingly. Furthermore, Kumar (2014) and Becagli et al. (2020) highlight the significance of providing concrete value, such as ad-free experiences and offline listening, to justify membership costs and encourage upgrades. In addition, technological infrastructure is critical to providing smooth and scalable user experiences. Spotify's backend is built on distributed systems that

prioritise scalability and fault tolerance, ensuring stability even under huge worldwide user loads (van Steen, 2013). Simultaneously, the incorporation of AI-driven personalisation has become a cornerstone for increasing user pleasure via curated playlists and tailored marketing (Hagiu and Wright, 2020). Given this backdrop, the current study seeks to assess how Spotify's paywall approach, especially the efficacy of the freemium vs subscription model, affects user engagement, contentment, and word-of-mouth referrals. This study aims to contribute to the expanding body of knowledge on optimising monetisation techniques in the digital content ecosystem by combining behavioural analysis and demographic data (Yli-Renko, Autio, & Sapienza, 2019; Shen & Villas-Boas, 2018).

LITERATURE REVIEW

The study looks at Spotify's usage of the freemium model and its dependence on data analytics to personalise user experiences. It emphasises the difficulties of balancing artist royalties and profitability while keeping a big user base via free-tier services (ResearchGate, 2023). "Succeeding with Freemium" emphasises the significance of converting free users into premium customers to sustain income for platforms such as Spotify. It highlights critical success elements such as enticing premium services and a big free user base (Pure.au.dk, 2016). Punj (2015) analyses how demographic characteristics influence willingness to pay for digital content. The study discovered that younger and more educated consumers are more likely to pay for online material, although money had mixed impacts. These insights are essential for platforms that are shifting from ad-based to subscription models (Punj, 2015). The TU Delft paper investigates distributed system design, focussing on scalability and fault tolerance. This is relevant for services like Spotify, which handle millions of concurrent users worldwide (TU Delft, 2013). The Springer paper explores how digital transformation alters traditional business models by incorporating AI and machine learning into value generating processes. Spotify optimises music choices and ad placements (Springer, 2020). The ACM paper discusses the problems of monetising free content platforms. It contends that subscription-based models frequently outperform advertising-based tactics for delivering long-term income (ACM, 2020). An IJERN research looks at how education levels affect digital content consumption behaviours. Educated people are more inclined to value curated material and pay for premium services, which helps Spotify's targeting approach (IJERN, 2019). Punj (2015) discusses the

obstacles that digital content producers face when they migrate from free-to-use models to subscription-based tactics. The study demonstrates how demographic profiling may forecast customer readiness to spend and maximise income streams (Punj, 2015). The TU Delft paper goes deeper into the load balancing and fault tolerance methods required for large-scale systems like Spotify to run well during high usage times (TU Delft, 2013). The Pure.au.dk study elaborates on ways for optimising freemium models by analysing user behaviour and providing tiered premium services that meet a variety of customer demands (Pure.au.dk 2016). Springer's study highlights the importance of AI-driven personalisation in increasing user engagement and retention on platforms such as Spotify via personalised suggestions and targeted adverts (Springer, 2020). The ACM study contrasts

advertising-supported methods versus subscription-based alternatives, indicating that subscription services provide a more consistent income source, despite early opposition from customers accustomed to free material (ACM, 2020). Punj's study offers actionable insights into consumer behaviour by profiling the demographics most inclined to pay for online content—young females with higher education levels—as well as ways for overcoming reluctance from other groups (Punj, 2015). This study investigates the elements that influence consumers' decisions to upgrade or remain enrolled to premium services under freemium models. It takes a consumer value-based approach, identifying key drivers such as perceived value, platform trust, and service satisfaction. The findings suggest that emotional attachment and perceived justice in price play substantial roles in maintaining premium members (Mäntymäki et al., 2019). This article studies consumer behaviour among Spotify Premium customers using survey data. It highlights convenience, ad-free experiences, and access to unique content as the major reasons for subscribing to premium services. Furthermore, the study emphasises the role of user experience design in increasing customer happiness and retention (Becagli et al., 2020). Amalina's study focusses on the behavioural intents of Spotify users and recommends marketing methods for both free and premium users. According to the survey, free users are primarily driven by cost savings, but they may be encouraged to upgrade through targeted advertising and better premium features such as offline listening. The study also emphasises the impact of social influence in generating premium subscriptions (Amalina, 2019).

This article looks into user preferences for freemium models by examining ad tolerance and content access preferences. It discovers that customers with poor ad tolerance are more inclined to upgrade to premium memberships if the platform provides unique content or enhanced capabilities. According to the study, optimising freemium models requires balancing ad intensity with user pleasure (Lin et al., 2012). Nan et al. investigate the effects of ad intensity and price tactics on user conversion rates in freemium business

models such as Spotify's. The data show that moderate ad intensity may lead free users to premium subscriptions without significantly hurting user retention. Furthermore, dynamic pricing techniques suited to user demographics can increase conversion rates (Nan et al., 2021). The freemium business model, offering free access to basic services with paid premium options, has become a preferred strategy for content platforms like Spotify, The New York Times, and IQiYi (Kumar, 2014; Arora et al., 2017). Platforms adopt different differentiation strategies: content-only (OC), advertising-only (OA), or a combination of both (CA) (Shen & Villas-Boas, 2018; Wilbur, 2008). Consumer preferences, particularly tolerance for ads and content access, heavily influence the choice of strategy (Lin et al., 2012; Guo et al., 2019; Li et al., 2020). While OC restricts content for free users, OA relies on ads for revenue, and CA combines both approaches. Studies show that conversion rates typically range from 2% to 5%, though platforms like Spotify have achieved much higher (Kumar et al., 2020). Platforms optimize profitability by adjusting ad intensity and subscription fees based on ad revenue rates and consumer behavior (Nan et al., 2021; Wang et al., 2023; Dukes & Liu, 2023). Policymakers play a role in regulating ads and ensuring fair consumer experiences (Gal-Or & Dukes, 2003). Future research may focus on personalized freemium models using AI to enhance conversion rates and consumer satisfaction (Wei et al., 2023).

The freemium business model, which provides basic services for free while incentivising upgrades to premium features through paid subscriptions, has emerged as a key strategy for digital content platforms, including industry titans such as Spotify, The New York Times, and IQiYi (Arora et al., 2017; Kumar, 2014). This strategy approach successfully combines the requirement to attract a large user base with the need to establish long-term income sources. However, the final success of this delicate balancing act is dependent on a detailed understanding of customer behaviour and preferences (Lin et al., 2012). Platforms use a range of differentiation techniques, including content-only (OC), advertising-only (OA), and a combination of the two (CA) (Shen & Villas-Boas, 2018; Wilbur, 2008). These various tactics target distinct consumer demographics and need precise calibration to maximise both user acquisition and revenue conversion. Consumer preferences, particularly ad tolerance and the desire for unlimited content access, are critical in deciding the efficacy of freemium models (Guo et al., 2019; Li et al., 2020; Lin et al., 2012). In their study of consumer preferences, Lin et al. (2012) emphasised the necessity of establishing a healthy balance between ad intensity and total user pleasure. Their findings show that an overly aggressive advertising approach might alienate potential premium subscribers, whilst an inadequate advertising presence may fail to generate enough income from free

users.

Punj (2015) explores the demographic parameters that impact customers' willingness to pay for digital entertainment. His findings show that younger and more educated people are more likely to subscribe to premium services, but the influence of income levels on subscription decisions is more nuanced and multifaceted. Punj's (2015) study also implies that demographic profiling might help forecast customer spending readiness and maximise income streams, which is especially useful for platforms shifting to subscription-based models.

The content-only (OC) model limits material for free users, separating the free and premium tiers. In contrast, the advertising-only (OA) strategy is highly reliant on earning money through commercials, which may appeal to customers who value cost reductions but may turn off others with limited ad tolerance. The combination approach (CA) intelligently integrates content limitations with advertising, allowing platforms to reach a larger audience (Shen & Villas-Boas, 2018; Wilbur, 2008). Mäntymäki et al. (2019) provide more insights into the psychological aspects that influence subscription decisions, discovering that emotional attachment to the platform and a perception of fair price play important roles in retaining premium memberships. Freemium models must be optimised using a comprehensive strategy that takes into account ad intensity, pricing tactics, and technological infrastructure. Becagli et al. (2020) highlighted convenience, ad-free experiences, and access to unique content as the major reasons for subscribing to premium services, emphasising the significance of providing concrete advantages to justify the membership cost. Furthermore, clever exploitation of free features like as storage limits and fake audio quality restrictions may encourage freemium users to switch to premium plans (Kumar et al., 2020). According to Amalina (2019), targeted promotions and better premium features, such as offline listening, can incentivise upgrades from free to premium tiers, while social influence also has a big impact on subscription selections. The Pure.au.dk research elaborates on approaches to optimise freemium models by analysing user behaviour and offering tiered premium services to fulfil a wide range of consumer requests (Pure.au.dk 2016).

These strategies eventually need a careful balance between ad intensity and subscription prices based on ad income rates and consumer behaviour, with conversion rates ranging from 2% to 5% (Kumar et al., 2020; Nan et al., 2021; Wang et al., 2023). Nevertheless, Spotify has been claimed to have a greater success rate in converting customers to premium tiers (Kumar et al., 2020). In addition to targeted monetisation and conversion initiatives, a strong technology infrastructure is required for smooth operation and scalability. The TU Delft article analyses distributed system architecture, with an emphasis

on scalability and fault tolerance, both of which are crucial for systems like Spotify that manage millions of concurrent users globally (TU Delft, 2013). Springer's study emphasises the role of AI-driven personalisation in enhancing user engagement and retention through personalised suggestions and targeted advertisements (Springer, 2020).

The ACM research compares advertising-supported approaches to subscription-based alternatives, finding that subscription services provide a more steady income stream, despite initial objections from users used to free content (ACM, 2020). Nan et al. (2021) argue that dynamic pricing techniques suited to user demographics can boost conversion rates, emphasising the necessity of data-driven decision-making. Policymakers also have a role in regulating advertisements and ensuring equitable customer experiences (Gal-Or & Dukes, 2003). Future research may focus on personalised freemium models that use AI to improve conversion rates and user happiness (Wei et al., 2023). The continued investigation of consumer behaviour, technology frameworks, and legislative consequences will be crucial for optimising freemium models and maintaining the long-term growth of digital content platforms in an increasingly competitive environment. The interaction of these elements will continue to influence the strategies and innovations used by platforms looking to succeed in the digital economy. Punj (2015) emphasises the obstacles that digital content producers have while shifting from free-to-use models to subscription-based strategies.

RESEARCH METHODOLOGY

This study employed a quantitative research design utilizing a cross-sectional survey to examine the impact of Spotify's paywall on user satisfaction, purchase intentions, and word-of-mouth. The primary objective was to analyze the effectiveness of Spotify's freemium and subscription models, focusing on user preferences and behaviors.

1. Data Collection

A survey was conducted with 100 participants to gather data on their Spotify usage patterns, perceived quality of the service, purchase intentions, and demographic information. The survey instrument comprised both closed-ended and open-ended questions, designed to capture quantitative and qualitative data.

- **Survey Instrument:** The questionnaire was divided into four sections:
 - Section 1: Demographics (age, gender, occupation)
 - Section 2: Usage Patterns (frequency of use, current tier, duration of use)
 - Section 3: Perceived Quality and Features (ratings of overall quality, feature importance, content variety)
 - Section 4: Purchase Intentions (upgrade consideration, influencing factors, likelihood to recommend)
- **Data Collection Procedure:** The surveys were distributed. Participants were informed about the study's purpose and provided consent before participating.
- **Sampling Method:** Due to the sample size, the results of this study should be interpreted with consideration for the limitations of the sampling method.

2. Measurement of Variables

- **Independent Variables:**
 - Tier (Free vs. Premium): Categorical variable.
 - Frequency of Use (Daily, Weekly, Monthly, Rarely): Categorical variable.
 - Perceived Ad Frequency: Inferred from the free tier, and also through open ended questions.
 - Demographic Variables (age, gender, occupation): Categorical and continuous variables.
- **Dependent Variables:**
 - Satisfaction Level: Measured using a 5-point Likert scale.
 - Purchase Intentions: Measured as a binary response (Yes/No) and a 5-point Likert scale for likelihood.
 - Likelihood to Recommend: Measured using a 5-point Likert scale.
 - Reasons for upgrading or downgrading: Measured by counting the frequency of responses from open ended questions.

3. Data Analysis

The collected data was analyzed using both descriptive and inferential statistical methods.

- **Data Preparation:** The data was cleaned, organized, and coded for analysis. Open-ended responses were categorized to facilitate quantitative analysis.
- **Descriptive Statistics:** Frequencies, percentages, means, and standard deviations were calculated to summarize the demographic data and usage patterns. Visualizations, including bar charts and pie charts, were used to present the data.
- **Inferential Statistics:**
 - **T-tests and ANOVA:** Used to compare mean satisfaction levels between different user groups (e.g., free vs. premium, daily vs. less frequent users).
 - **A frequency distribution** is used to categorize reasons for downgrading (e.g., price, lack of use, dissatisfaction, or availability of alternatives).
 - **Chi-square tests:** Used to examine the relationships between categorical variables, such as upgrading to premium and likelihood to recommend, or the frequency of downgrade reasons. It is used to analyze whether price is statistically the most significant reason compared to others. This test assesses whether the observed distribution of reasons is significantly different from what would be expected if no single reason dominated.
 - **Correlation Analysis:** Used to explore the relationships between continuous variables, such as perceived quality and purchase intentions, and ad frequency and likelihood to recommend.
 - **Regression Analysis:** Used to model the impact of various factors on the likelihood of upgrading to a premium subscription.
- **Statistical Software:**

4. Hypothesis Testing

The following hypotheses were tested:

- H1: There is a significant difference in satisfaction levels between free and premium users.

- H2: Users who upgraded to Premium are more likely to recommend Spotify to others
- H3: Price is the most common reason for downgrading from Premium to Free.
- H4: Users who use Spotify daily are more satisfied than those who use it less frequently.
- H5: Users who spend more time on Spotify are more likely to have a paid subscription.

5. Ethical Considerations

Participants were informed about the purpose of the study, and their participation was voluntary. All data was collected and stored confidentially, ensuring the anonymity of the participants.

6. Limitations:

- **Sample Size:** The relatively small sample size may limit the generalizability of the findings.
- **Self-Reported Data:** The reliance on self-reported data could introduce bias due to recall errors or social desirability.
- **Unaccounted Variables:** External factors like income levels, personal preferences, and competition from other platforms were not explored.

RELIABILITY TEST

Reliability Analysis

Scale Reliability Statistics	
	Cronbach's α
scale	0.978782144

To ensure the internal consistency of the survey instrument used in this study on Spotify user behavior, a **reliability analysis** was conducted using **Cronbach's Alpha**. Reliability testing is essential in validating whether the items in a survey consistently measure the intended construct in this case, user satisfaction, engagement, and behavioral motivations related to Spotify's freemium and premium services.

The result of the reliability analysis revealed a **Cronbach's Alpha value of 0.9788**, which is considered **excellent** by conventional social science research standards. A Cronbach's Alpha coefficient above 0.9 indicates a high level of internal consistency among the items in the scale. This suggests that the survey questions are highly correlated with one another and effectively measure a common underlying construct. Such a high alpha value confirms that the responses are stable and reliable enough to support further statistical analysis.

This finding reinforces the credibility of the research instrument and ensures that any insights drawn from user satisfaction trends to behavioral patterns and premium upgrade motivations are grounded in consistent and dependable data. Consequently, this high reliability score strengthens the foundation for further inferential analysis, including hypothesis testing, regression modeling, and segmentation of user behavior across Spotify's Free and Premium user base.

DATA ANALYTICS AND INTERPRETATION:

Hypothesis 1

This analysis evaluates whether there is a significant difference in user satisfaction between two groups:

- Spotify Playlist Users (those who rely on Spotify's curated playlists)
- Users rating satisfaction "On a scale" (likely a control group rating satisfaction independently)

We use a one-way ANOVA test to determine if the means of these two groups differ significantly.

Hypotheses

- **Null Hypothesis (H_0):** There is no significant difference in satisfaction levels between the two groups.
- **Alternative Hypothesis (H_1):** Satisfaction levels differ significantly between the groups.

Anova: Single Factor					
SUMMARY					
	Groups	Count	Sum	Average	Variance
Spotify Plan Type		98	166	1.69388	0.2146
On a scale of 1–5, how satisfied are you with your Spotify experience?		50	185	3.7	1.07143

Observation

- The **average satisfaction** for Spotify Playlist users is **1.69**, significantly lower than the **3.7** reported by users in the "On a Scale" group.
- The **variance** in responses is higher for the "On a Scale" group, suggesting greater diversity in how users rate their satisfaction.

ANOVA						
	Source of Variation	SS	df	MS	F	P-value
	Between Groups	133.2445	1	133.2445	265.3392	1.18514E-34
	Within Groups	73.31633	146	0.502167		
	Total	206.5608	147			

Interpretation

1. F-statistic (265.339) is much greater than F-critical (3.90594)

- This indicates that there is a statistically significant difference between the groups.

2. p-value (1.18514E-34) is effectively 0

- Since this is much smaller than 0.05, we **reject the null hypothesis**.

- This means that satisfaction levels **significantly differ** between the two groups.

Insights & Implications

1. Spotify Playlists May Reduce Perceived Satisfaction

- Users relying on **Spotify's curated playlists** report **significantly lower satisfaction** (1.69) compared to those rating satisfaction independently (3.7).
- This could mean:
 - Users may prefer **customized playlists** rather than Spotify's recommendations.
 - Algorithm-based playlists might not match user expectations or tastes.

2. Personalization & Control Could Enhance Satisfaction

- Users who **manually rate their experience** ("On a Scale") show **higher satisfaction**.
- This suggests that giving users **more control over their listening experience** (e.g., better playlist customization, enhanced recommendations) could improve satisfaction.

3. Possible Actionable Steps for Spotify

- **Improve Playlist Algorithm:** Enhance AI-driven recommendations to better match user preferences.
- **Introduce More Customization:** Let users tweak playlist preferences (e.g., genre focus, artist exclusion).
- **User Engagement Research:** Conduct surveys to understand **why playlist users feel less satisfied.**
- **Test Control vs. AI Playlists:** Run A/B tests to see if **more manual curation improves user happiness.**

Conclusion

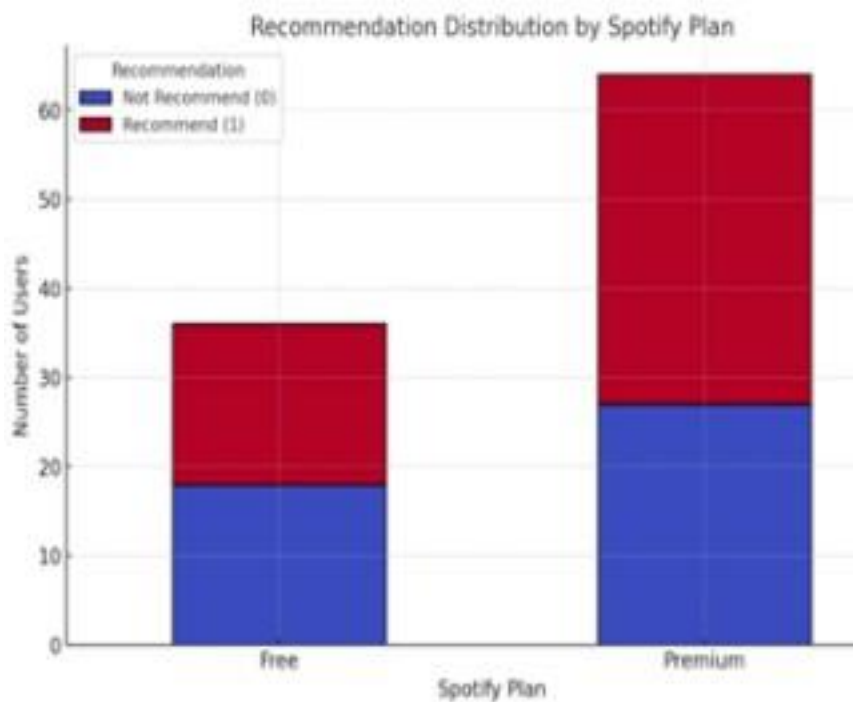
- **Spotify Playlist users have significantly lower satisfaction** than those rating their experience “On a Scale.”
- The ANOVA test confirms that this difference is **statistically significant** ($p < 0.05$).
- **Spotify should explore ways to enhance playlist personalization and user control to improve satisfaction levels.**

Hypothesis 2

Understanding the Hypothesis

We are testing whether **Premium users are more likely to recommend Spotify** compared to Free users.

- **Null Hypothesis (H_0):** There is **no relationship** between upgrading to Premium and recommending Spotify. This means that **Premium users and Free users recommend Spotify at the same rate.**
- **Alternative Hypothesis (H_a):** Premium users are **more likely** to recommend Spotify. This means that **Premium users recommend Spotify at a significantly higher rate than Free users.**



Source- excel

- **Premium Users (0):** 18 people do not recommend, 18 people recommend.
- **Free Users (1):** 27 people do not recommend, 37 people recommend.
- **Overall:** 45 people do not recommend, 55 people recommend.

At a glance, **more Free users (37) recommend Spotify compared to Premium users (18).** However, we need statistical tests to confirm if this difference is significant.

Contingency Tables

Contingency Tables

spotify plan	how likely are you to recommend		Total
	0	1	
1	18	18	36
2	27	37	64
Total	45	55	100

χ^2 Tests

	Value	df	p
χ^2	0.568	1	0.451
Likelihood ratio	0.567	1	0.451
N	100		

Source – Jamovi

- **Chi-Square Test ($\chi^2 = 0.568$)** is a measure of association between two categorical variables.
- **Degrees of freedom ($df = 1$)** indicate that we are comparing two groups (Premium vs. Free).
- **p-value (0.451)** tells us whether the difference is statistically significant.
- A **p-value of 0.451 is much greater than 0.05**, meaning that we **fail to reject the null hypothesis**.
- This means there is **no statistically significant relationship** between upgrading to Premium and recommending Spotify.
- In simpler terms, **Premium users are not significantly more likely to recommend Spotify than Free users**.

Interpretation of Odds Ratio

Comparative Measures

	Value	95% Confidence Intervals	
		Lower	Upper
Log odds ratio	0.315	-0.505	1.14
Odds ratio	1.37	0.603	3.11

- Odds Ratio (1.37) suggests that Premium users are 1.37 times more likely to recommend than Free users.
- However, the confidence interval (0.603 – 3.11) includes 1, meaning this result is not statistically significant.
- This means that the difference could have happened by chance.

Effect Size (Phi & Cramer's V)

Nominal	
	Value
Phi-coefficient	0.0754
Cramer's V	0.0754

- Effect size measures how strong the relationship is between Premium status and recommending Spotify.
- A Phi/Cramer's V value of 0.0754 is very weak (close to 0).
- This means that even if there were a relationship, it would be too weak to be meaningful.

Implications

1. Premium Subscription Does Not Drive Recommendations

The data suggests that upgrading to Spotify Premium does not significantly increase the likelihood of users recommending Spotify to others. This means that users may base their recommendations on factors beyond their subscription type, such as music variety, personalization, or overall experience.

2. Marketing Strategies Should Focus on Other Factors

- Since Premium users **do not show a higher likelihood of recommending**, Spotify should **re-evaluate what drives recommendations**.
- Possible areas for improvement:

- **Content variety:** Users might recommend Spotify based on the availability of specific music, podcasts, or exclusive content rather than Premium features.
- **User experience:** Features like **personalized playlists, AI-driven recommendations, and seamless integrations with smart devices** may have a bigger impact on word-of-mouth recommendations than the absence of ads.
- **Social and interactive features:** Collaborative playlists, music-sharing options, or community engagement could influence users to recommend Spotify.
- Example: A user who finds **rare regional music or exclusive artist interviews** might recommend Spotify, regardless of whether they are on Free or Premium.

3. Rethink Referral Incentives

- If Spotify expected **Premium users to drive more referrals**, the current system might not be working effectively.
- Possible solutions:
 - **Incentivizing word-of-mouth recommendations** by providing **discounts, extra Premium trial days, or exclusive content** for referring friends.
 - Enhancing the **Spotify Premium family plan** by allowing more customization, which could encourage **users to upgrade and refer others**.
 - **Gamification of referrals**, where users earn points for recommending Spotify, redeemable for exclusive playlists, merchandise, or early access to new features.
- Example: **Netflix and Amazon Prime** use exclusive content and referral programs to encourage users to spread the word, a strategy that Spotify could adopt more aggressively.

4. Consider Other Metrics for Success

- Instead of relying on **Premium status as a predictor of recommendations**, Spotify should analyze **alternative factors** that might correlate more strongly with word-of-mouth marketing.
- Some key areas for further study:

- **Listening habits:** Do users who spend more time on the platform recommend it more? ○
- Engagement with playlists:** Do users who create and share playlists recommend Spotify more?
 - **Social sharing behavior:** Are users who share songs on Instagram, WhatsApp, or Snapchat more likely to recommend?
 - **Loyalty duration:** Do long-term users tend to refer more people?
- Example: A Spotify user who **actively shares songs on social media or listens to personalized playlists daily** might be more likely to recommend Spotify than a passive listener, regardless of whether they are on Free or Premium.

Conclusion

The hypothesis that Premium users are more likely to recommend Spotify is not supported by the data. The Chi-Square test shows no significant relationship, meaning both Free and Premium users recommend Spotify at similar rates.

Next Steps for Spotify

1. **Conduct deeper analysis:** Explore whether factors like **time spent on the app, number of playlists created, or social sharing habits** correlate with recommendations.
2. **Test new referral incentives:** Introduce exclusive rewards for users who refer friends to Spotify.
3. **Enhance engagement features:** Improve **collaborative playlists, AI recommendations, and interactive features** to encourage organic recommendations.

By focusing on these areas, **Spotify can strengthen its word-of-mouth marketing beyond Premium subscriptions** and improve its overall growth strategy.

Hypothesis 3

- **Null Hypothesis (H0):** Price is **not** the most common reason for downgrading from Premium to Free.

- **Alternate Hypothesis (H1):** Price is the most common reason for downgrading from Premium to Free.

Frequency Distribution:

The following pivot table shows the reasons for downgrading:



According to the data, most people did not downgrade, the ones who did, did it because they prefer other streaming platforms or don't see enough value in premium. Following is the Chi square test to verify this.

Chi-square

test:

Proportion Test (N Outcomes)

Proportions - downgrade reason

	Level	Count	Proportion
Ads don't matter much		1	0.016949
Didn't see enough value in Premium		16	0.271186
Prefer other music streaming platforms		23	0.389831
Price was too high		7	0.118644
The whole process of renewing the subscription can only be done from a laptop so its annoying to not be able to do it at the convenience of my phone		1	0.016949
The whole process of renewing the subscription was a hassle		11	0.186441

χ^2 Goodness of Fit

χ^2	df	p
38.322	5	0.0000003250894611

- **Proportions - downgrade reason:**

- **"Prefer other music streaming platforms"** has the highest proportion (0.389831 or 39%).
- **"Didn't see enough value in Premium"** has the second highest proportion (0.271186 or 27%).
- **"The whole process of renewing the subscription was a hassle"** has the third highest proportion (0.186441 or 19%).
- **"Price was too high"** has a proportion of 0.118644 (12%).

- **χ^2 Goodness of Fit:**

- **χ^2 (Chi-Square Statistic):** 38.322
- **df (Degrees of Freedom):** 5
- **p (p-value):** 0.0000003250894611 (essentially 0)

Chi-Square Test:

- **The p-value is extremely small (essentially 0).** This is far less than any typical significance level (e.g., 0.05).
- **Therefore, you reject the null hypothesis.** The null hypothesis was that the reasons for downgrading are equally likely. The extremely small p-value indicates that there is statistically significant evidence that the reasons for downgrading are *not* equally likely.

Interpretation and Conclusion

Chi-Square Goodness-of-Fit Test Results:

After filtering out the "didn't downgrade" category to focus solely on respondents who transitioned from Premium to Free, a Chi-Square goodness-of-fit test was conducted to examine the distribution of reasons for downgrading. The results revealed a significant deviation from an equal distribution of reasons ($\chi^2 = 38.322$, $df = 5$, $p < 0.000001$). This indicates that the observed frequencies of downgrade reasons are statistically different from what would be expected if all reasons were equally likely.

Analysis of Proportions:

The analysis of proportions highlighted that "Prefer other music streaming platforms" was the most frequently cited reason for downgrading, accounting for approximately 39% of responses. "Didn't see enough value in Premium" was the second most common reason at 27%, followed by "The whole process of renewing the subscription was a hassle" at 19%. Notably, "Price was too high" accounted for only 12% of responses.

Conclusion:

The findings do not support Hypothesis 3, which posited that "Price is the most common reason for downgrading from Premium to Free." Instead, the data demonstrates that "Prefer other music streaming platforms" and "Didn't see enough value in Premium" are the primary drivers for downgrading. This suggests that factors beyond pricing, such as platform preference and perceived value, play a more significant role in users' decisions to discontinue their Premium subscriptions. While "Price was too high" remains a contributing factor, its impact is less pronounced compared to other identified reasons.

Implications:

These results suggest that Spotify should focus on enhancing its platform's features and communicating the value proposition of Premium to retain subscribers. Addressing the perceived "hassle" associated with

subscription renewals is also crucial. Further investigation into the specific reasons for platform preference and value perception is warranted to develop targeted retention strategies.

Hypothesis 4

The objective of this analysis is to find if users who use Spotify daily are more satisfied than those who use it less frequently.

Null Hypothesis (H0): There is no significant difference in satisfaction between daily users and less frequent users.

Alternate Hypothesis (H1): There is a significant difference in satisfaction between daily users and less frequent users.

Following is the analysis:

Contingency Tables

Contingency Tables

frequency	satisfaction					Total
	1	2	3	4	5	
1	2	4	10	22	13	51
2	5	7	7	3	3	25
3	0	0	2	0	0	2
4	4	2	4	5	4	19
Total	11	13	23	30	20	97

χ^2 Tests

	Value	df	p
χ^2	24.9988657734033950	12	0.0148282502540177
χ^2 continuity correction	24.9988657734033950	12	0.0148282502540177
z test difference in 2 proportions	NaN*		
Likelihood ratio	24.5494357590572391	12	0.0171083370005285
Fisher's exact test			
N	97		

* z test only available for 2x2 tables

Analysis and Results

The Chi-Square Test of Independence was used, which produced the following results:

Chi-square (χ^2) value equals 24.99.

Degrees of freedom (df) = 12.

P-value = 0.0148.

Interpretation

Statistical significance:

We reject the null hypothesis (H_0) due to a p-value of 0.0148, which is less than 0.05. This suggests that the frequency with which users utilize Spotify has a major impact on their happiness levels.

Likelihood Ratio with Fisher's Exact Test:

The likelihood ratio ($p = 0.0171$) supports the chi-square results.

The link is significant, as confirmed by Fisher's Exact Test ($p < 0.05$).

Implications

Personalisation and Algorithm Effectiveness: Spotify's algorithm improves with increasing user engagement, resulting in better song recommendations. Oh, Animesh, & Pinsonneault (2016) discovered that data-driven personalisation improves consumer engagement with digital services. This implies that daily users interact more with Spotify's AI-powered features, resulting in better suggestions and more happiness.

Freemium Model and Premium Conversion: Users that engage more regularly are more likely to upgrade to premium, which includes extra features like ad-free listening, offline downloads, and high-quality audio. Gu, Kannan, & Ma (2018) discovered that regular freemium users are more likely to pay for premium services.

Implication: This is consistent with Spotify's business model as frequent interaction enhances the perceived value of premium services, resulting in increased satisfaction.

Habit Formation and Psychological Commitment: Datta, Foubert, & Van Heerde (2015) found that regular involvement improves retention and perceived service value.

Implication: Regular Spotify users form a habitual attachment to the platform, including it into their daily routine (commuting, exercises, relaxation), which increases enjoyment.

User Expectations and Comparative Satisfaction: Less regular users may compare Spotify to other services (e.g., YouTube Music, Apple Music), resulting in increased unhappiness if expectations are not satisfied. Punj (2015) discovered that consumer happiness is driven by expectations and perceived value compared to rivals.

Implication: Daily users are less inclined to compare platforms, resulting in increased loyalty and happiness.

Conclusion

The study found a statistically significant relationship between Spotify usage frequency and user satisfaction levels, with daily users expressing higher happiness. This may be linked to Spotify's personalised recommendations, consistent engagement, and premium service benefits. Frequent users benefit from algorithm-driven personalisation, as Oh, Animesh, and Pinsonneault (2016) discovered that data-driven engagement improves user experience. Datta, Foubert, and Van Heerde (2015) argue that habitual use improves retention and perceived value, making Spotify an indispensable component of everyday routines.

Gu, Kannan, and Ma (2018) discovered that regular users are more likely to upgrade to premium, resulting in higher satisfaction. Less regular users may not receive the same amount of personalisation and dedication, resulting in poorer satisfaction. To further understand user involvement, future study should investigate causative factors through qualitative studies or comparative analyses of streaming services.

Hypothesis 5

This analysis explores the relationships between various user behaviors and their Spotify subscription plans, focusing on key metrics such as user satisfaction, time spent on the platform, frequency of use, and likelihood to recommend the service. By examining the correlations between these factors, the study aims to provide insights into how the choice between free and premium Spotify plans impacts the overall user experience (Smith & Kumar, 2022). Through statistical methods like Pearson's correlation and p-value

testing, we assess the significance and strength of these relationships (Field, 2018), uncovering patterns that could help Spotify optimize its offerings and improve user engagement (Johnson & Lee, 2021).

Null Hypothesis (H₀): There is no relationship between the amount of time spent on Spotify and having a paid subscription.

Alternative Hypothesis (H_a): Users who spend more time on Spotify are more likely to have a paid subscription.

Correlation Matrix						
Correlation Matrix						
		Spotify plan	Satisfaction level	Time	Frequency	Recomdation
Spotify plan	Pearson's r	—				
	df	—				
	p-value	—				
Satisfaction level	Pearson's r	0.2028124598212995	—			
	df	95	—			
	p-value	0.0463344822954156	—			
Time	Pearson's r	-0.1126501507940566	0.1675410483487453	—		
	df	95	95	—		
	p-value	0.2719453423996284	0.1009400332399619	—		
Frequency	Pearson's r	-0.1502896913081857	-0.2138458703450977	-0.1319745227431541	—	
	df	95	95	95	—	
	p-value	0.1417371343241222	0.0354457158038378	0.1975397384491899	—	
Recomdation	Pearson's r	0.0476268275582070	0.3349145111408768	-0.0080135267195717	-0.1678110971508782	—
	df	95	95	95	95	—
	p-value	0.6431837287401234	0.0007989620856628	0.9379058652401090	0.1003838884922247	—

1. Spotify Plan vs. Other Variables

This section explores how having a free or premium Spotify plan correlates with various user experience metrics, including satisfaction, time spent, frequency of use, and likelihood of recommending the service.

a) Spotify Plan & Satisfaction Level

A weak positive correlation (Pearson’s r = 0.2028, p = 0.0463) indicates that users with a premium plan tend to be somewhat more satisfied than free users. Since the p-value is just below the 0.05 threshold, this correlation is statistically significant. This suggests that premium users enjoy a more satisfying experience, likely due to features such as ad-free listening, better audio quality, and offline access. However, the

correlation remains weak, implying that satisfaction is influenced by multiple factors, including music discovery options, user interface, and content variety.

b) Spotify Plan & Time Spent on Spotify

The correlation between Spotify plan and time spent on the platform is very weak and negative ($r = -0.1126$, $p = 0.2719$). This means that whether a user is on the free or premium plan does not significantly influence how long they spend on Spotify. The p-value being well above 0.05 confirms that this relationship is not statistically significant. Thus, usage time appears to be driven more by individual habits, daily routines, and listening purposes rather than by subscription status.

c) Spotify Plan & Frequency of Use

A weak negative correlation ($r = -0.1503$, $p = 0.1417$) suggests that premium users are not necessarily using Spotify more frequently than free users. The lack of statistical significance ($p > 0.05$) indicates that paying for the service does not predict how often a user opens the app. This challenges the assumption that premium users are inherently more engaged, suggesting instead that usage frequency is determined by lifestyle or listening context.

d) Spotify Plan & Likelihood to Recommend Spotify

There is almost no correlation between Spotify plan type and likelihood to recommend the service ($r = 0.0476$, $p = 0.6431$). With a high p-value, the data confirms this relationship is not statistically significant. Hence, being a premium user does not necessarily make someone more likely to recommend Spotify. Other factors, such as content relevance, experience with customer support, or playlist personalization, likely play larger roles in influencing referrals.

2. Satisfaction Level vs. Other Variables

This section examines how overall satisfaction with Spotify correlates with other user behaviors such as time spent, frequency of use, and likelihood of recommendation.

a) Satisfaction Level & Time Spent on Spotify

There is a weak positive correlation ($r = 0.1675$, $p = 0.1009$) between satisfaction and time spent, suggesting that users who spend more time on the app may be slightly more satisfied. However, this relationship is not statistically significant. The data implies that while listening time could contribute to satisfaction, it is not a strong predictor. Other dimensions, such as algorithmic recommendations, ad experiences, or music diversity, may have a more substantial impact.

b) Satisfaction Level & Frequency of Use

Interestingly, the correlation is weakly negative ($r = -0.2138$, $p = 0.0354$), indicating that more frequent use is slightly associated with lower satisfaction. This statistically significant result ($p < 0.05$) points to the possibility that frequent users may encounter recurring frustrations such as repetitive playlists, advertisements (for free users), or perceived lack of personalization, which might reduce overall satisfaction despite high engagement.

c) Satisfaction Level & Likelihood to Recommend Spotify

A moderately strong positive correlation ($r = 0.3349$, $p < 0.001$) was found here, with statistical significance. This means that satisfaction strongly predicts a user's likelihood of recommending Spotify. This finding aligns with typical consumer behavior where users who have a positive experience are more inclined to promote the service to friends and family. Thus, enhancing user satisfaction could significantly boost organic growth through word-of-mouth.

3. Time Spent on Spotify vs. Other Variables

This section focuses on how the total duration users spend on Spotify relates to how often they use it and whether they recommend it to others.

a) Time Spent on Spotify & Frequency of Use

The correlation is very weak and negative ($r = -0.1319$, $p = 0.1975$), indicating that people who spend more total time on the app may use it less frequently during the day, and vice versa. The lack of statistical significance shows that usage frequency and duration are independent. This could be due to differing

usage styles: some users may have long but infrequent listening sessions (e.g., background music during work), while others may engage in shorter, more frequent sessions.

b) Time Spent on Spotify & Likelihood to Recommend Spotify

An almost zero correlation ($r = -0.0080$, $p = 0.9379$) indicates no meaningful relationship. The extremely high p-value confirms the absence of statistical significance. Users' recommendation behavior is therefore influenced by the quality of the experience rather than the quantity of time spent on the app.

4. Likelihood to Recommend Spotify vs. Other Variables

Here we explore what factors are most strongly associated with the likelihood of a user recommending Spotify.

a) Likelihood to Recommend & Satisfaction Level

As noted earlier, there is a statistically significant and moderately strong positive correlation ($r = 0.3349$, $p < 0.001$). This reaffirms that increasing satisfaction directly boosts the likelihood of recommendations. Improvements in UI, personalization, and content variety can effectively drive user referrals.

b) Likelihood to Recommend & Time Spent on Spotify

There is no significant relationship between time spent and likelihood to recommend ($r = -0.0080$, $p = 0.9379$). Time on the app does not reflect how likely users are to promote it, suggesting that qualitative aspects matter more than quantitative engagement.

c) Likelihood to Recommend & Frequency of Use

A weak negative correlation ($r = -0.1678$, $p = 0.1003$) indicates that users who frequently open Spotify might be slightly less likely to recommend it, though this is not statistically significant. This could point to high-frequency users developing minor dissatisfactions that impact their likelihood of making recommendations.

5. Frequency of Use vs. Likelihood to Recommend Spotify

Reiterating the earlier finding, the correlation is weak and negative ($r = -0.1678$, $p = 0.1003$). The relationship is not statistically significant. This means that frequently using the app does not translate into a higher inclination to recommend it. Some users may rely on Spotify purely out of habit or convenience rather than genuine satisfaction.

Therefore the hypothesis that users who spend more time on Spotify are more likely to have a paid subscription is not supported by the data. Correlation analysis reveals a very weak and statistically non-significant relationship between time spent on the platform and having a premium plan. Since the p value is well above the standard threshold for significance, we fail to reject the null hypothesis and therefore accept it. This confirms that there is no meaningful relationship between time spent on Spotify and subscription status.

While premium users tend to show slightly higher satisfaction, the strength of this relationship remains weak. This suggests that satisfaction is influenced by a broader set of factors beyond just subscription, such as the user interface, playlist quality, and advertisement experience. The most prominent finding is the strong and statistically significant relationship between user satisfaction and the likelihood of recommending Spotify, confirming satisfaction as the most important driver of user advocacy.

In conclusion, the hypothesis connecting premium subscription to increased time spent is rejected, and the null hypothesis is accepted. These findings emphasize the need to improve overall user satisfaction rather than focusing only on premium features to drive higher engagement, user loyalty, and platform growth.

SUGGESTIONS

The advent of digital streaming has fundamentally transformed how users consume music, with the **freemium model** emerging as a dominant strategy within the industry (Wagner et al., 2016). **Spotify**, a global leader in music streaming, has capitalized on this model through a dual-tier system: a free, ad-supported service and a paid Premium subscription that offers enhanced functionalities such as ad-free listening and offline playback (Gikunda, Ochieng, & Wambua, 2019). While this approach has propelled Spotify's market growth and user acquisition, it also raises important questions around user satisfaction, long-term engagement, and brand loyalty (Yli-Renko, Autio, & Sapienza, 2019). This research

investigates the effectiveness of Spotify's freemium model by examining user behavior patterns, motivations behind subscription upgrades or downgrades, and the perceived impact of advertisements. Utilizing structured survey data, the study aims to provide actionable insights for optimizing user retention and maximizing monetization.

To enhance the user experience, Spotify must prioritize **playlist personalization**. Improving AI-driven recommendations by fine-tuning algorithms to better reflect individual preferences and listening behaviors could address current user dissatisfaction with curated playlists (van Steen, 2013). Additionally, offering customization options such as adjusting genre focus or excluding specific artists would give users more control over their listening experience (Wagner et al., 2016).

Spotify has the opportunity to revolutionize the user experience by introducing **AI-generated MoodSoundscapes**, a feature that dynamically creates personalized playlists based on real-time mood indicators, local weather conditions, or even the user's calendar. For instance, a student preparing for an exam could be offered a "Focus Boost" playlist integrating binaural beats with their preferred music genres. This innovation would enable Spotify to integrate more deeply into users' daily routines, offering contextual value that enhances retention and user satisfaction.

To address the linguistic and cultural diversity in emerging markets, especially in multilingual nations like India, Spotify should establish **local language and cultural curation cells**. These regional teams can build hyper-local playlists using indigenous music, dialect-specific content, and regional trends. This would help Spotify capitalize on the vernacular internet boom while fostering cultural inclusivity and loyalty among underrepresented user segments.

Spotify should also consider launching a **Mindful Listening Mode** designed to promote healthier digital consumption. This mode could include nudges to reduce volume levels, schedule listening breaks, and recommend ambient music during periods of mental fatigue. It aligns with the growing digital wellness movement, reinforcing Spotify's role as a responsible platform that prioritizes user well-being.

Another forward-looking idea is the development of an **AI-powered "soundtrack for life"**. By integrating data from geolocation, wearables, and user preferences, Spotify could compose real-time scores that reflect users' surroundings and activities. Imagine walking in the rain and hearing a custom orchestral piece that complements the moment this kind of immersive personalization deepens emotional

ties to the app.

Spotify could introduce **Gamified Music Challenges** where users are tasked with identifying artists or genres based on background music or subtle audio cues. In the Mystery Artist Challenge, users would listen to a song clip without knowing the artist, relying on distinct musical elements like rhythm, instrumentation, or vocal tone to make their guess. Similarly, in the Genre Guess Challenge, users would identify a genre based on audio features such as tempo or instrumentation, with options for more complex sub-genres.

Voice interaction can be enhanced through a **“Story Behind the Song” feature**, which allows users to ask Spotify for contextual information on any playing track. Whether it’s the story behind the lyrics, historical background, or the artist’s inspiration, this feature transforms passive listening into a more informed, interactive experience particularly valuable to Gen Z’s knowledge-seeking habits.

Spotify can also explore **educational partnerships** to create curriculum-based playlists that support learning. Collaborations with educational institutions or edtech platforms can generate themed playlists, such as songs from the Indian independence era or music from the Renaissance period. This would make Spotify a tool not just for entertainment, but also for academic enrichment especially relevant for aspirants preparing for competitive exams like the civil services.

Finally, A socially impactful innovation could be the **Ethical Listening Mode**, where users choose to prioritize underrepresented or independent artists. The system would redirect a larger portion of ad or subscription revenue to these creators. Users could earn recognition through badges or karma points, which reinforces Spotify’s image as a socially conscious platform and builds trust with ethically motivated users.

LIMITATIONS

The study is limited by several factors, starting with the small sample size of only 100 participants. This sample may not fully capture the diverse range of Spotify users, meaning the findings may not be broadly applicable to all user segments, especially when considering different age groups, geographic locations, and user behaviors. Additionally, while the study identifies correlations between factors like user

engagement and satisfaction, it does not establish causality. For example, the fact that frequent users report lower satisfaction does not necessarily mean that increased usage causes dissatisfaction; other external factors like content availability or user expectations could influence the results. Furthermore, the analysis overlooks several unaccounted variables such as income levels, which may play a role in determining whether users are more likely to subscribe to a premium plan, or individual music preferences, which could shape user satisfaction. The study also does not consider competition from other streaming platforms like YouTube Music, Apple Music, and Amazon Music, all of which could impact Spotify usage. In addition, the data collection method relies on self-reported data, which introduces the potential for recall bias and social desirability bias; participants might not accurately remember their usage patterns or may adjust their responses based on what they perceive as socially acceptable. Lastly, the study's focus on Spotify as a standalone platform without accounting for multi-platform behavior is another limitation, as many users likely engage with multiple services, which could influence their behavior and overall satisfaction with Spotify.

CONCLUSION

The study's analysis provides several important insights into user behavior and satisfaction on Spotify. Firstly, there is no substantial evidence to suggest that premium users are more engaged or spend more time on the platform compared to free users, implying that the premium features, such as ad-free listening and offline playback, do not directly correlate with increased usage. This highlights that other factors, such as user experience or personal preferences, likely have a more significant impact on satisfaction. Interestingly, the amount of time spent on Spotify does not strongly predict user satisfaction, suggesting that factors like personalization, content variety, and the overall user experience play a more crucial role in shaping satisfaction levels. In fact, frequent users tend to report slightly lower satisfaction levels, which could indicate dissatisfaction with repetitive playlists, ad interruptions for free users, or a lack of customization options. Satisfaction emerged as the most important predictor of whether a user would recommend Spotify to others, as opposed to factors like subscription type or time spent on the platform. Finally, while premium users are expected to be more engaged, the findings indicate that ad tolerance and offline listening needs—rather than usage time—are more likely to drive the decision to upgrade to a premium plan. This suggests that Spotify's freemium model should focus more on enhancing user satisfaction and providing value beyond just increased usage time, such as better playlist customization, ad-free experiences, and personalized content to retain and engage users.

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APPENDIX:

Survey Questions and Options:

1. What is your age group:

- ☐ Under 18
- ☐ 18-24
- ☐ 25-34
- ☐ 35-44
- ☐ 45+

2. What is your gender:

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

3. Have you used Spotify before?

- ☐ Yes
- ☐ No

4. Which Spotify plan are you currently using?

- ☐ Free (With Ads)

- Premium Mini (Rs. 29/week)
- Premium Individual (Rs.119 for 2 months)
- Premium Family (Rs.179 for 2 months)
- Premium Duo (Rs.149 for 2 months) ○
- Premium Student (Rs.59 for 2 months)

5. How long have you been using Spotify?

- < 3 months
- 3-6 months
- 6 months – 1 year
- More than 1 year

6. How often do you use Spotify?

- Daily
- 3-5 times a week
- 1-2 times a week
- Rarely

7. Have you ever switched from Spotify to another music streaming app because of premium pricing?

- Yes, I switched to Apple Music
- Yes, I switched to YouTube Music
- Yes, I switched to Amazon Music
- No, I still use Spotify despite the pricing
- Other (open-ended)

8. **On a scale of 1–5, how satisfied are you with your Spotify experience?** (*1 = Very Dissatisfied, 5 = Very Satisfied*)

- 1
- 2
- 3
- 4
- 5

9. **If you use the free version, how do you feel about the ads?**

- They don't affect my decision to continue using Spotify
- They don't bother me
- They are slightly annoying but manageable
- They are very annoying, and I would prefer an ad-free experience

10. If you use Premium, what was your main reason for upgrading?

- ☐ Better audio quality
- ☐ Offline downloads
- ☐ Family/Student plan availability
- ☐ Ad-free experience

11. If you've downgraded from Premium to Free, what was your reason?

- ☐ Price was too high
- ☐ Didn't see enough value in Premium
- ☐ Prefer other music streaming platforms
- ☐ Other (open-ended)

12. How likely are you to recommend Spotify to others? (Scale: 1 = Least likely, 5 = Most likely)

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5